ELEC 205 Design Assignment 2
Sequential Circuit Design

Due by 30 April 2026

Problem Statement

Design a **synchronous sequential circuit** with:

- One binary input: X
- Two binary outputs: Z_1 and Z_0
- A clock input

The circuit monitors the input stream formed by the input X at each clock cycle and operates as follows:

- Detect the specific sequence: $0 \to 1 \to 0$
  - When detected, set $Z_1 = 1$ and $Z_0 = 0$ for exactly one clock cycle.
- Detect the alternate sequence: $0 \to 1 \to 1$
  - When detected, set $Z_1 = 0$ and $Z_0 = 1$ for exactly one clock cycle.
- In all other cases, $Z_1 = 0$ and $Z_0 = 0$.

After any detection, the circuit must **immediately restart detection** from the current input. Overlap between patterns are not allowed.

Example Input Sequence:

$$(X) = (0),\quad (1),\quad (0),\quad (1),\quad (1),\quad (0),\quad (1),\quad (1),\quad (0),\quad (1),\quad (0)\ \ldots$$

Corresponding Output Sequence:

$$(Z_1, Z_0) = (0,0),\ (0,0),\ (0,0),\ (1,0),\ (0,0),\ (0,0),\ (0,0),\ (0,0),\ (0,1),\ (0,0),\ (0,0),\ (1,0)\ \ldots$$

Tasks

1. Draw the **state diagram** representing the pattern detection using a Moore model.
2. Assign **binary codes to the states** (state encoding).
3. Construct the **state transition table**.
4. Design the circuit using a minimum number of **D flip-flops** and **NAND gates**. Complements of the inputs and state variables are available. Avoid using extra **NOT gates**.
5. Draw the **complete logic circuit diagram**, showing flip-flops, combinational logic, and outputs Z_1 and Z_0.
6. Implement your design using **Logisim**. Test I/O characteristics with input sequence in 20 consecutive clock cycles (test cases should examplify occurance of non-zero outpus at least two times in the test sequence). Ensure test cases yield the expected outputs. Report on the correctness of the outputs and include a screenshot of the timing diagram in your report.

Tips:

- In Logisim, flip-flops have asynchronous Set and Reset inputs. Ensure these inputs are pulled low (0) unless used intentionally.
- You can use the "Clock" generator from the "Wiring" tab in Logisim.
- Under Simulate → Timing Diagram, browse and investigate the timing diagram analysis by changing the inputs of your circuit and observing the outputs.

Submission

Submit the following files:

- Your report in PDF format, documenting your design and screenshots of your Logisim circuit.
- Your complete Logisim circuit file (.circ).